



Differentially Private Topic Modelling

Mapping Privacy Risks & Prototyping Secure Algorithms

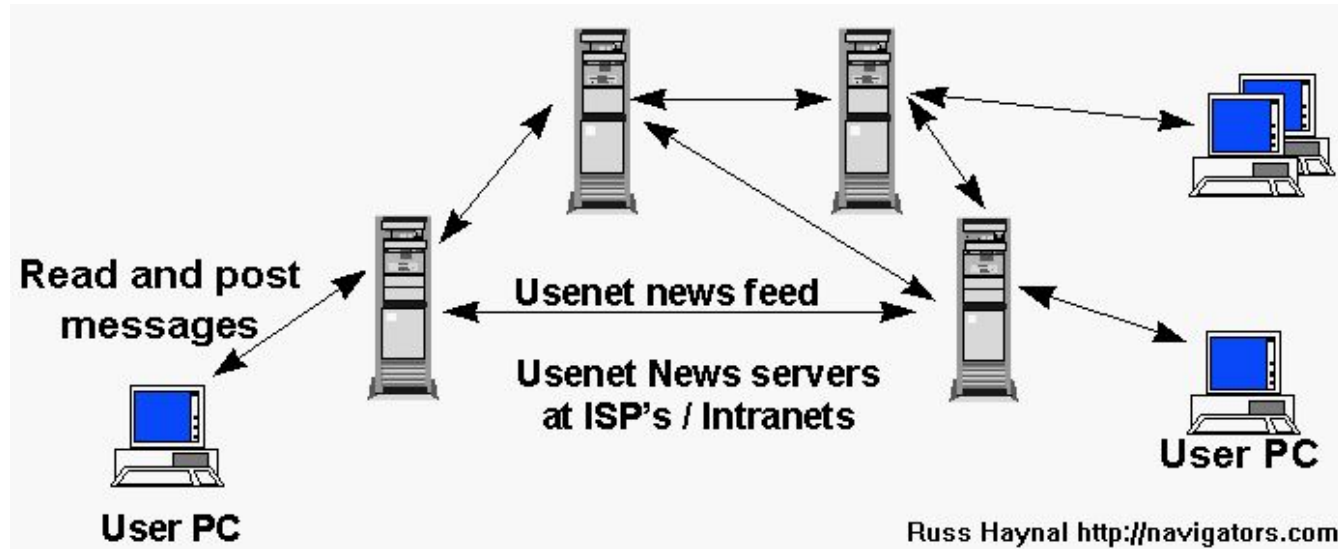
PPOL 6801 - **Muhammad Saad**

Project Overview

- **Latent Dirichlet Allocation (LDA)** used for text data applications including NLP
- **Uncover latent themes within a corpus** (*unsupervised*); documents as a mixture of topics inferred from their word distributions
- Are there **any privacy risks** associated with LDA?
- Can we implement **any privacy preserving techniques**?

Data Overview(I)

20 Newsgroup text dataset



Discussion groups on topic-specific forums called newsgroups
Decentralized microblogging platform before FB, Twitter, etc (early 1990s)

Data Overview(II)

Useful from a **privacy perspective**

Rows: 18,846

Columns: 5

```
$ text      <chr> "From: Mamatha Devineni Ratnam <mr47+@andrew.cmu.edu>\nSubject: Pens fans reactions...
$ group     <chr> "rec.sport.hockey", "comp.sys.ibm.pc.hardware", "talk.politics.mideast", "comp.sys...
$ subject   <chr> "Pens fans reactions", "Which high-performance VLB video card?", "Re: ARMENIA SAYS ...
$ from      <chr> "Mamatha Devineni Ratnam <mr47+@andrew.cmu.edu>", "mblawson@midway.ecn.uoknor.edu (...
$ email     <chr> "mr47+@andrew.cmu.edu", "mblawson@midway.ecn.uoknor.edu", "hilmi-er@dsv.su.se", "gu...
```

Includes **personal identifiers**

Privacy risks in text-based machine learning models

How can we have a publicly available text dataset with identifiers??

Apparently, identifiers are ‘*mostly obsolete*’ - ChatGPT

Data Overview(III)

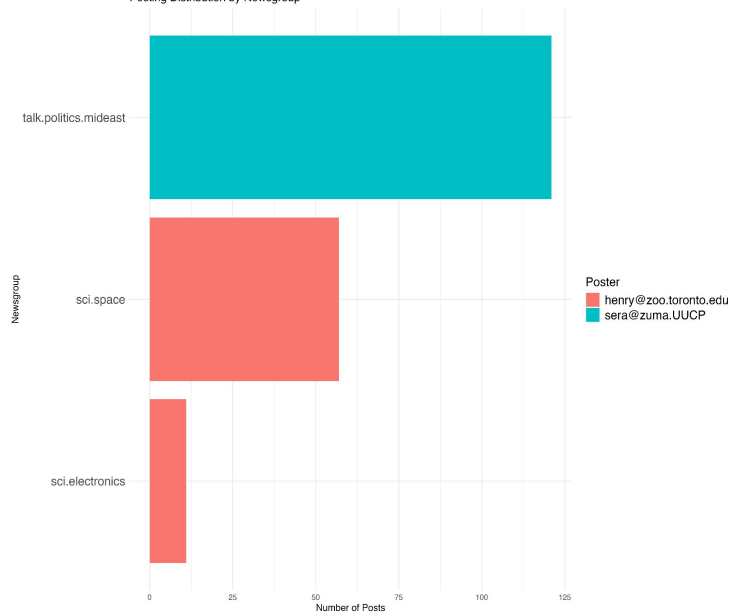
Total Documents: 18846

Users with Identifiable Emails: 8518

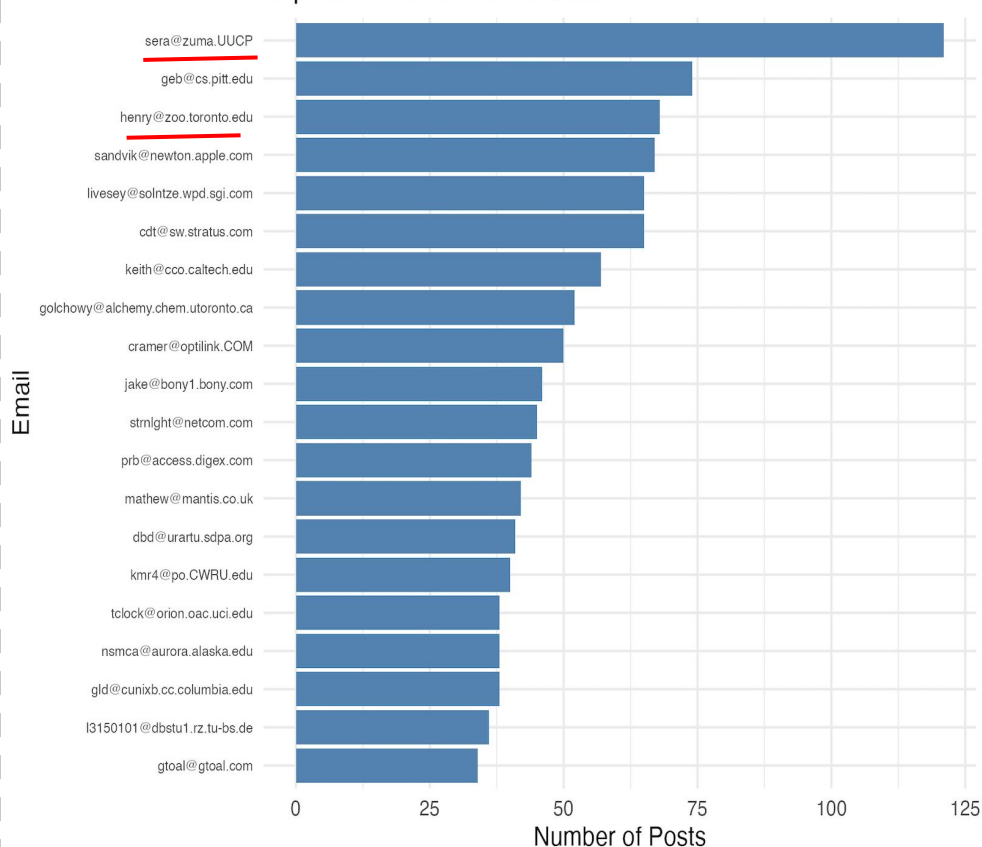
Max Posts by One User: 121

Median Posts per User: 1

Posting Distribution by Newsgroup



Top 20 Most Active Posters



Privacy Risks (I)

Google searched two of the top posters

sera@zuma.UUCP



ODTÜ Middle East Technical University
METU Research Information System



Prof. AHMET COŞAR

Faculty of Engineering, Department of Computer Engineering 

WoS Research Areas: Computer Science, Engineering Computing & Technology (Eng)

Avesis Research Areas: Information Systems, Communication and Control Engineering, Communication Engineering, Computer Sciences, Engineering and Technology

Figure [source](#).



‘*Serdar Argic*’ - spambot

henry@zoo.toronto.edu

Henry Spencer



Spencer is a Canadian programmer and space enthusiast who created three widely used, adapted, and influential regular expression libraries. In 1986, he was the first to release a regex library which could be freely included in other programs. Perl 2's regex package was based on and enhanced from Spencer's library, but Spencer's technological *tour de force* was creating the regex package used by Tcl. This implementation, Jeffrey Friedl writes, "is a hybrid [NFA/DFA engine] with the best of both worlds".

Websites: [Wikipedia](#), [O'Reilly bio](#), [Lysator](#), [Bio at NASA](#) (*photo source*)

Figure [source](#).

Privacy Risks (II)

Posts by top posters

sera@zuma.UUCP

‘Muslim women and children were massacred by the Armenians.’

‘Nazi Armenian Philosophy: Race above everything and before everything.’

henry@zoo.toronto.edu

‘We're repairing something with a 74ACT00 on it and the question arises, "well, do i really need the ACT part?"’

‘That is new stuff for me. So it means that you just can not put a satellite around around the Moon for too long’

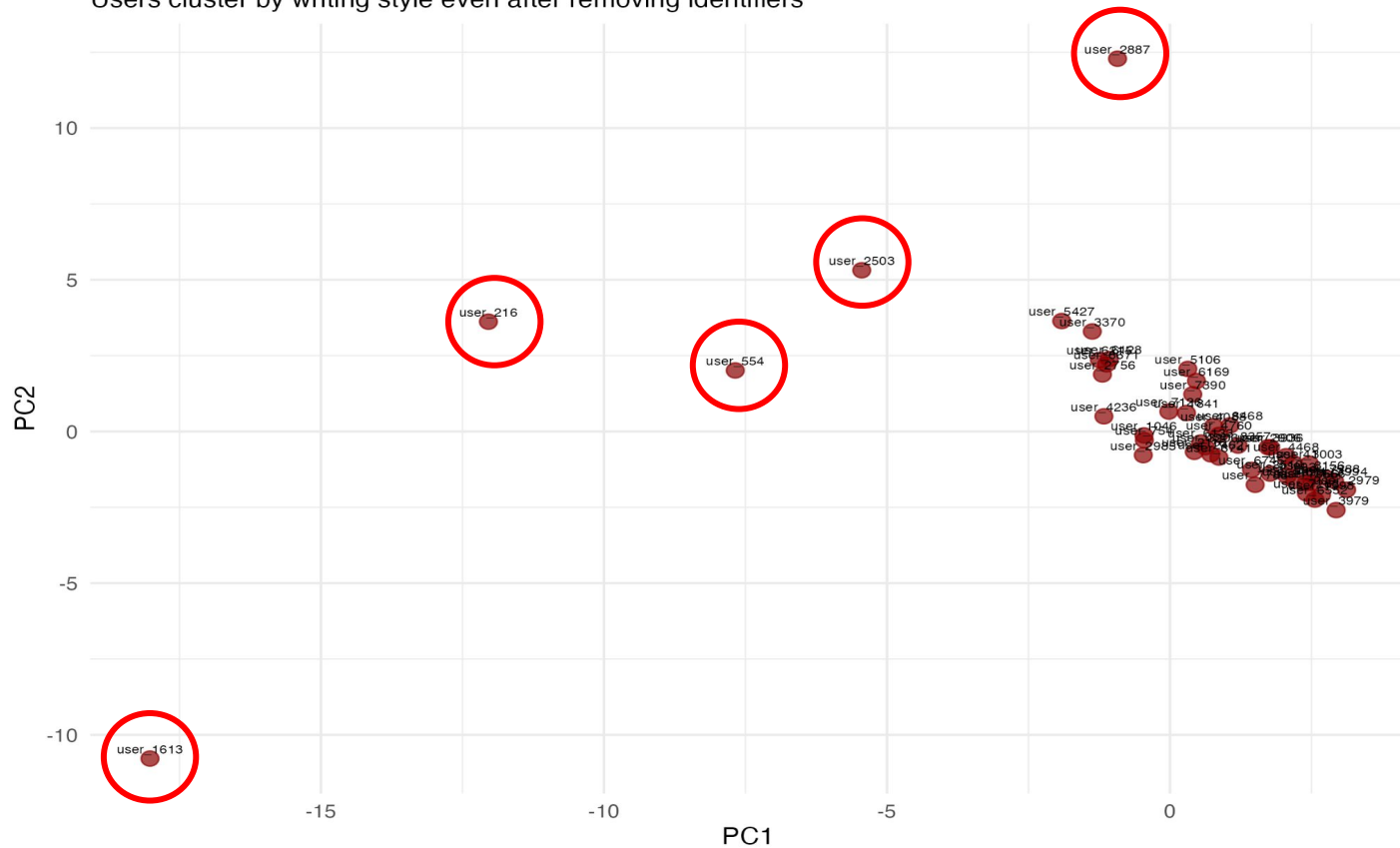
Each user leaves a unique, potentially identifiable digital fingerprint!

Pseudonymization Risks

Risk of re-identification based on unique stylistic contributions to text data (digital fingerprint)

Stylometric Fingerprinting After Pseudonymization

Users cluster by writing style even after removing identifiers



What about Topic Modelling?

Latent Dirichlet Allocation (LDA) topic modelling technique

Identify **structure of documents**; assign each document a **distribution of labels**

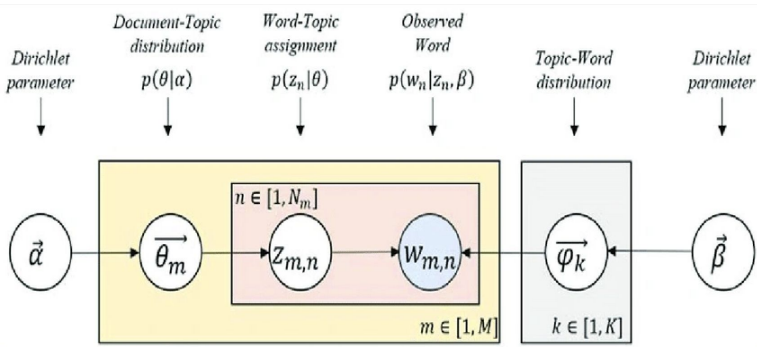


Figure [source](#)

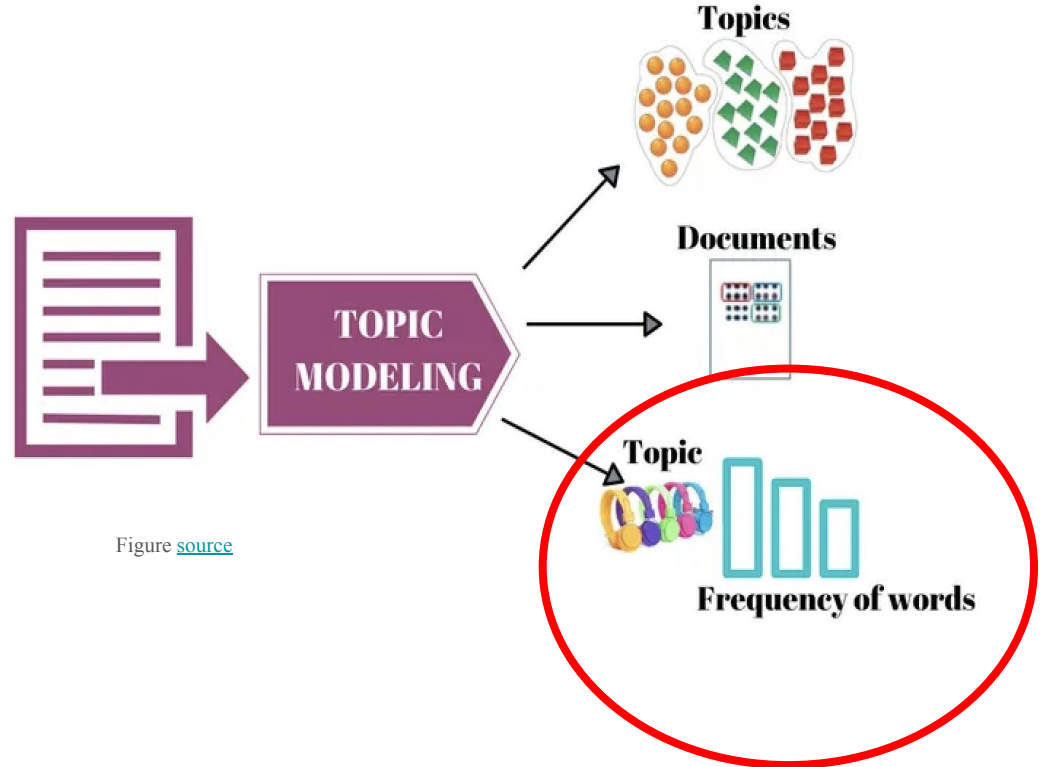


Figure [source](#)

20 Newsgroup LDA

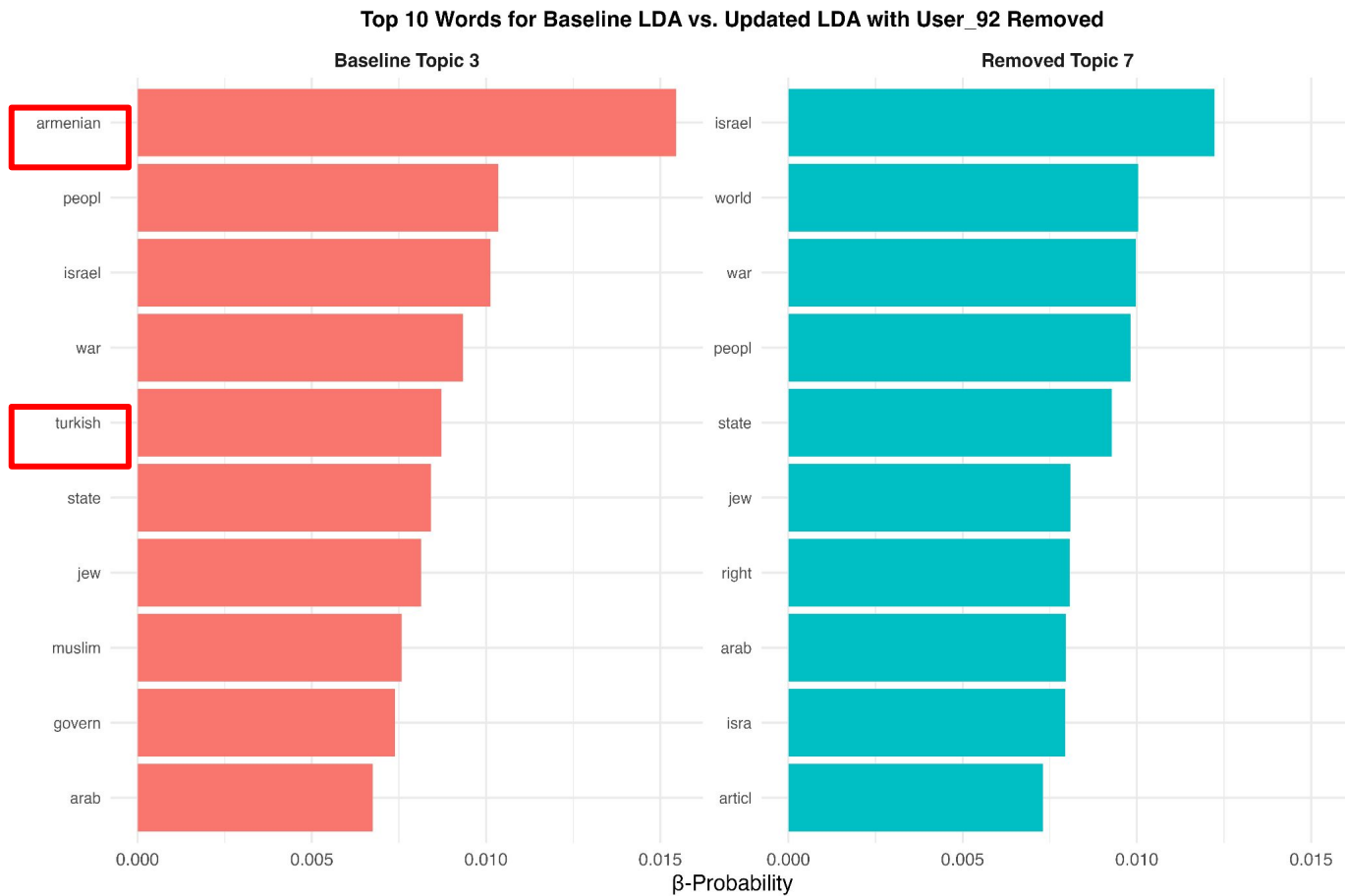
Ran LDA based on Gibbs sampling; $k = 20$ to mimic original groups

Top terms per LDA topic clearly map onto the original 20 Newsgroup categories



20 Newsgroup LDA - Memorization Risk (I)

If one user's data is removed, especially if that user is an outlier, it significantly affects LDA outputs, which increases privacy risks

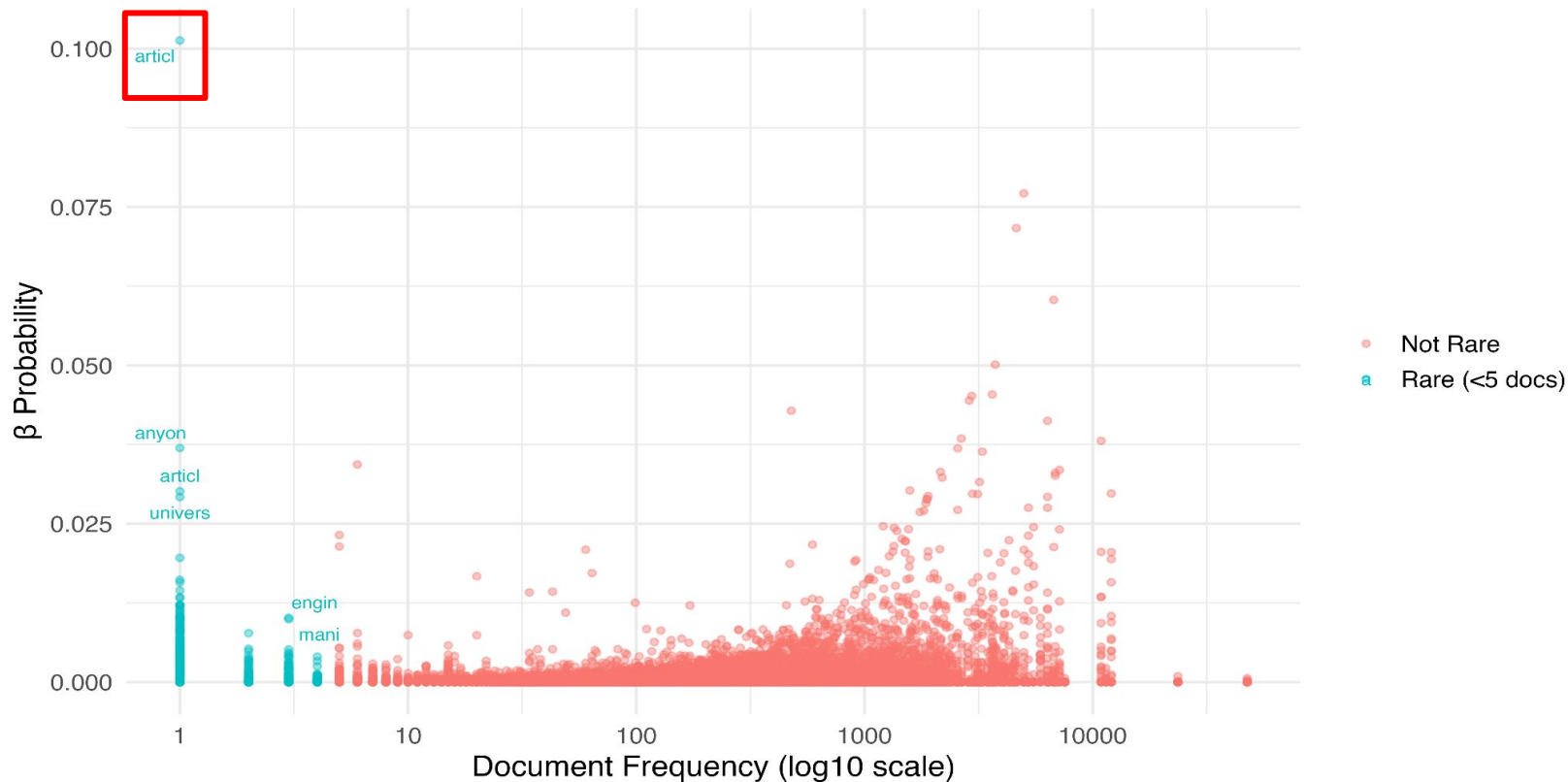


20 Newsgroup LDA - Memorization Risk (II)

An attacker who only sees the model outputs can infer rare term exists in the training corpus; strongly tied to a particular topic, possibly **one particular contributor (singling out!)**

Memorization Risk: Rare Words With High Topic Probabilities

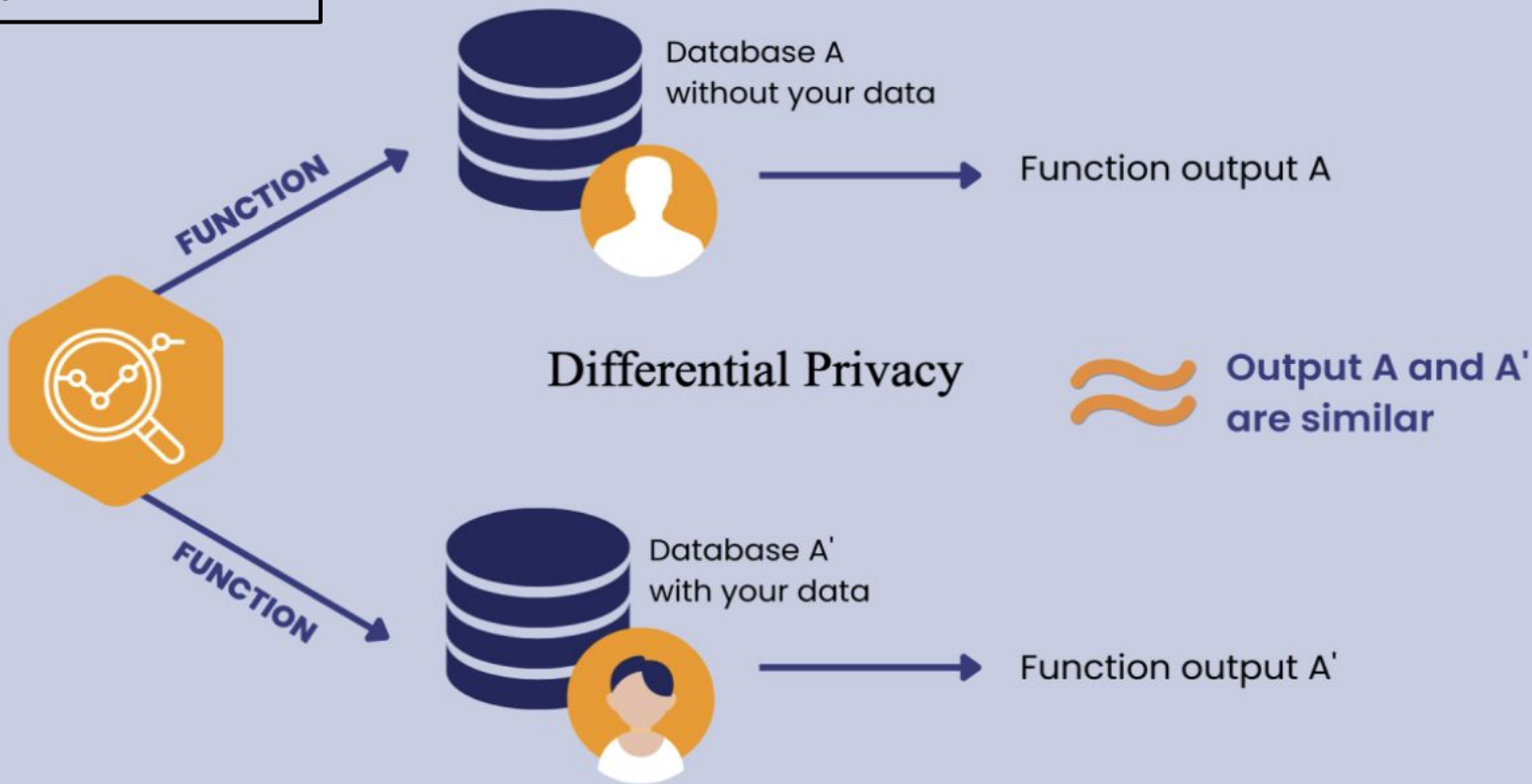
Rare terms (blue) appearing in <5 documents but assigned high β



Differential Privacy

Mathematical definition of privacy; helps us induce calibrated noise in statistical outputs to mask individual contributions

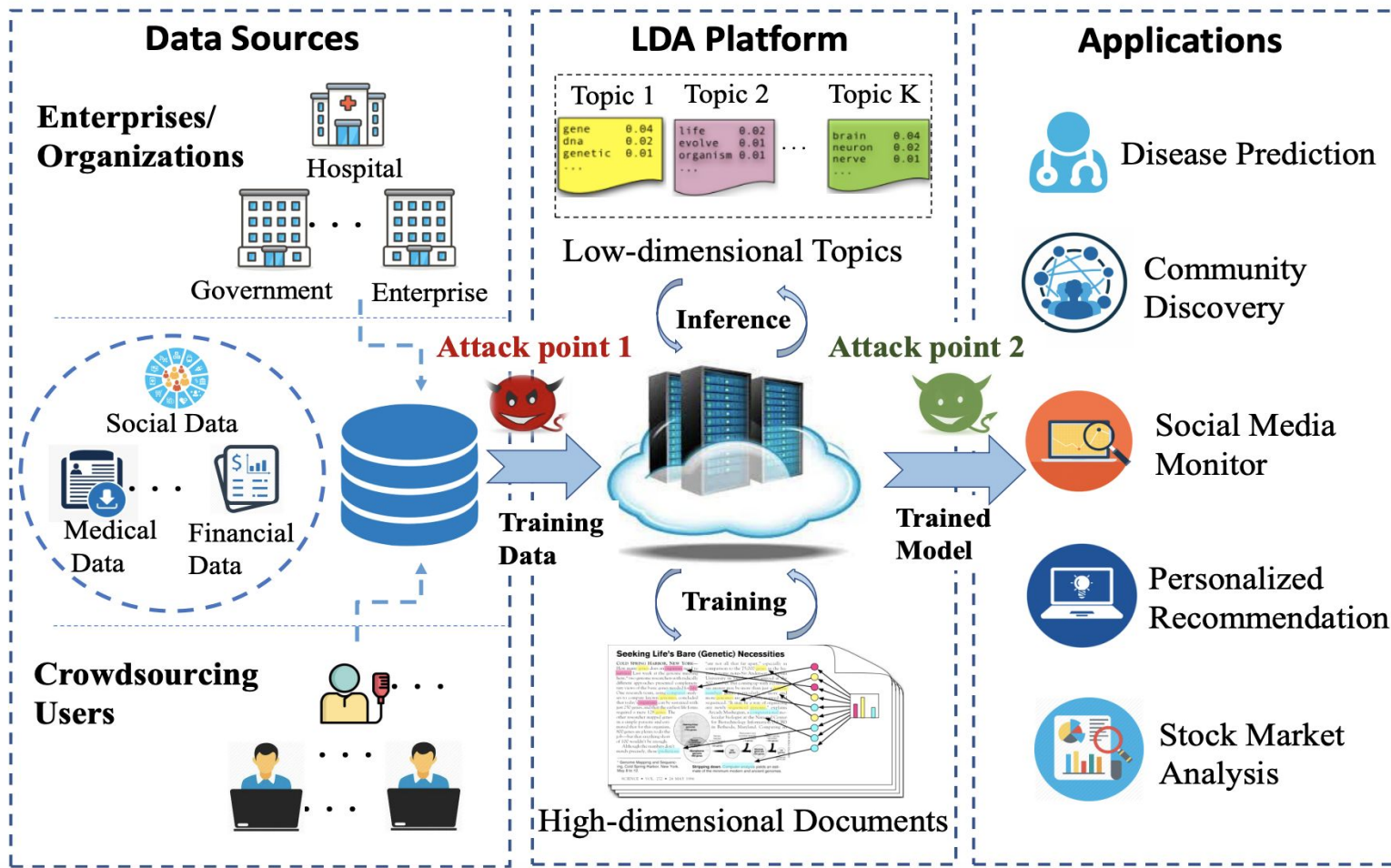
Figure [source](#)



Privacy Risks in LDA

Two attack points: *training*; *model output*. These are the two points where we can inject noise as well.

Figure [source](#)



Differentially Private LDA - Project Approach

Multiple points where you can inject calibrated noise to perturb statistical outputs and mask individual contributions

Authors (Year)	Method Feature	Privacy Level / Mechanism
Zhao et al. (2020)	DP-LDA via Collapsed Gibbs Sampling	Adds Laplace noise to word–topic counts at each iteration; word-level, Laplace mechanism
Huang & Chen (2021)	Subsampled Laplace DP (SUB-LDA)	Poisson subsampling before noise addition; word-level, Laplace + subsampling
Huang et al. (2022)	Rényi-DP LDA	Replaces Laplace with Gaussian mechanism; word-level, Gaussian mechanism
Manzonelli et al. (2024)	Vocabulary-level DP (DP Set Union)	DP applied to vocabulary construction + training; vocabulary-level, DP Set Union
Saad (2025)	Statistics-Perturbation DP-LDA	Laplace noise added to topic–word (n_{kw}) and document–topic (n_{dk}) counts; count-level, Laplace mechanism

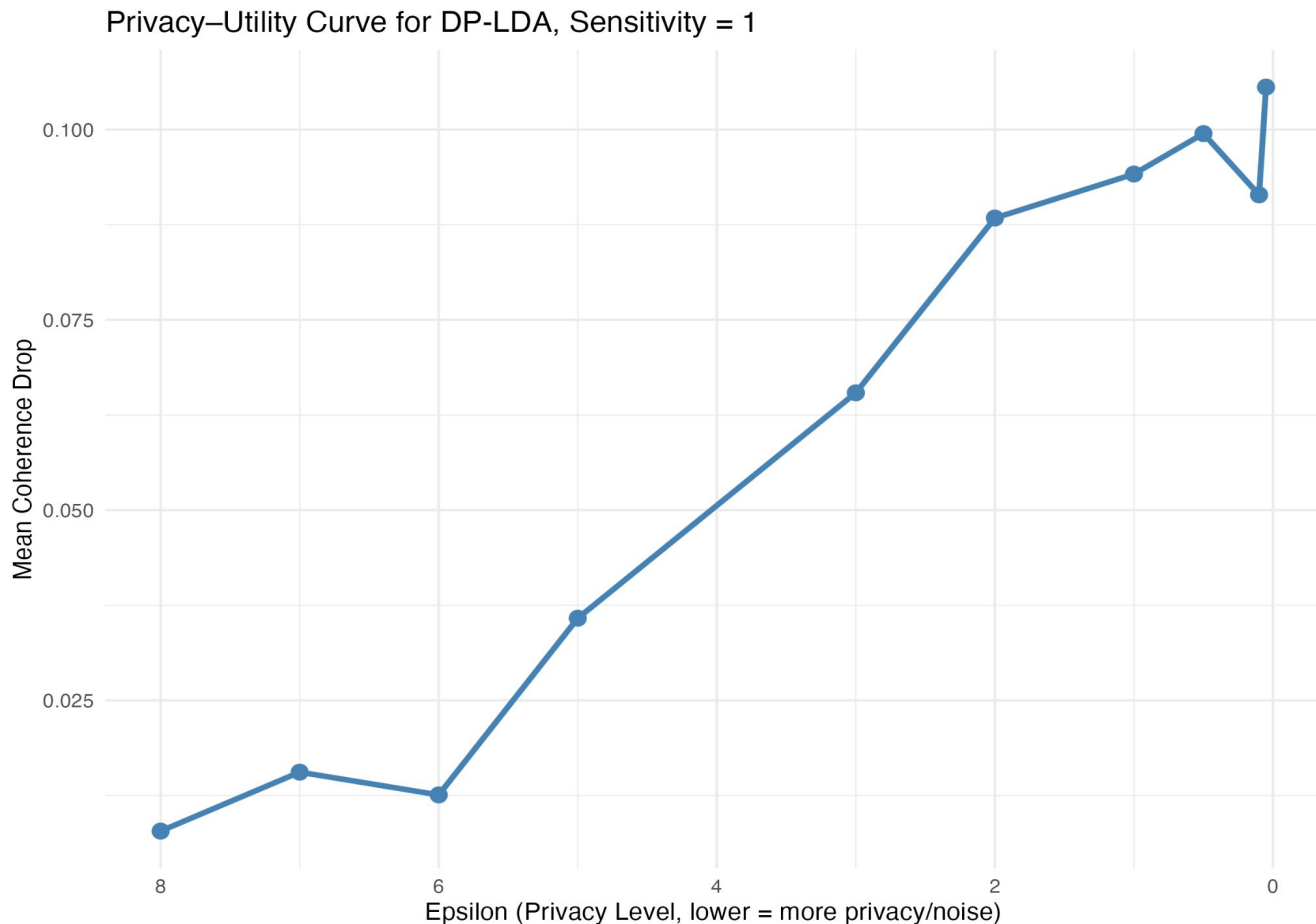
Injected noise directly into the *document–topic* + *topic–word* counts

Offered the most straightforward way to achieve differential privacy in Gibbs LDA

Simplistic application while still providing privacy guarantees

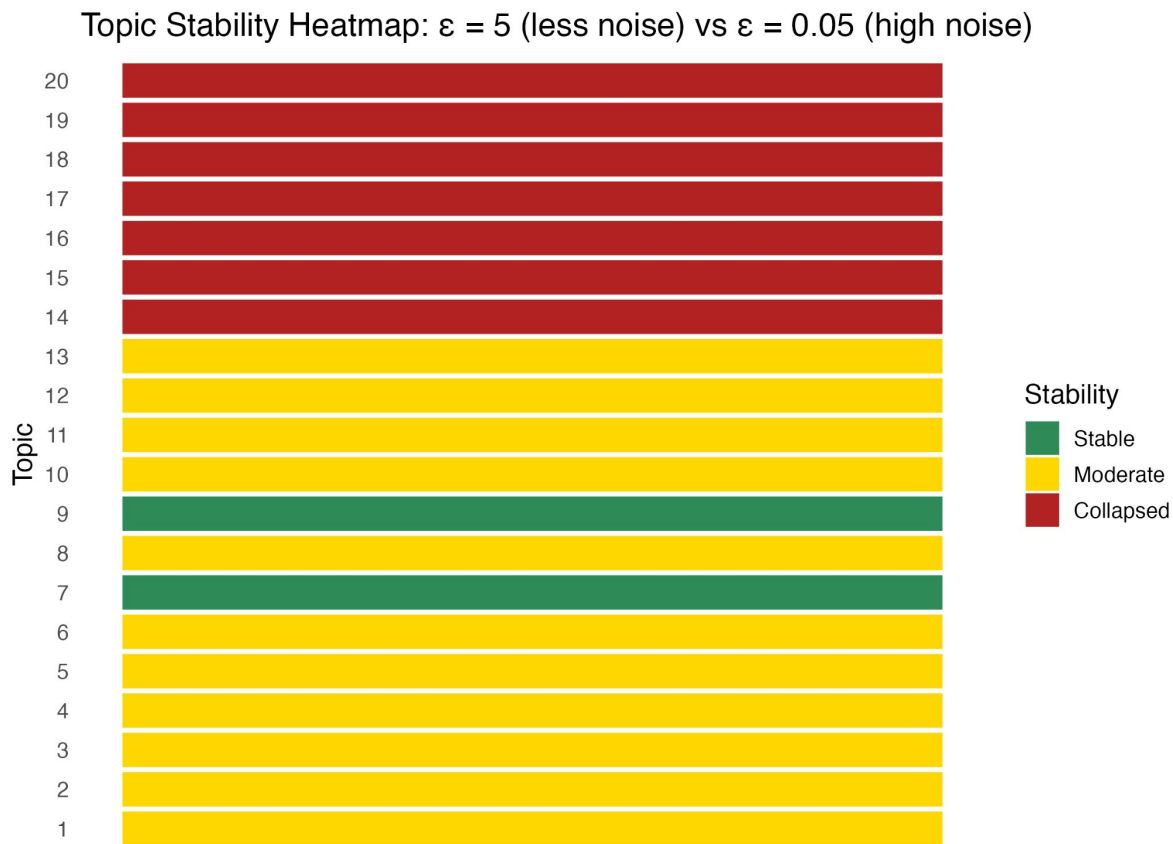
DP-LDA: Project Approach (Document-Level) (I)

Assume that addition/removal on one document in corpus can only change the topic-word count by one.



DP-LDA: Project Approach (Document-Level) (II)

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DP-LDA: Project Approach (Document-Level) (II)

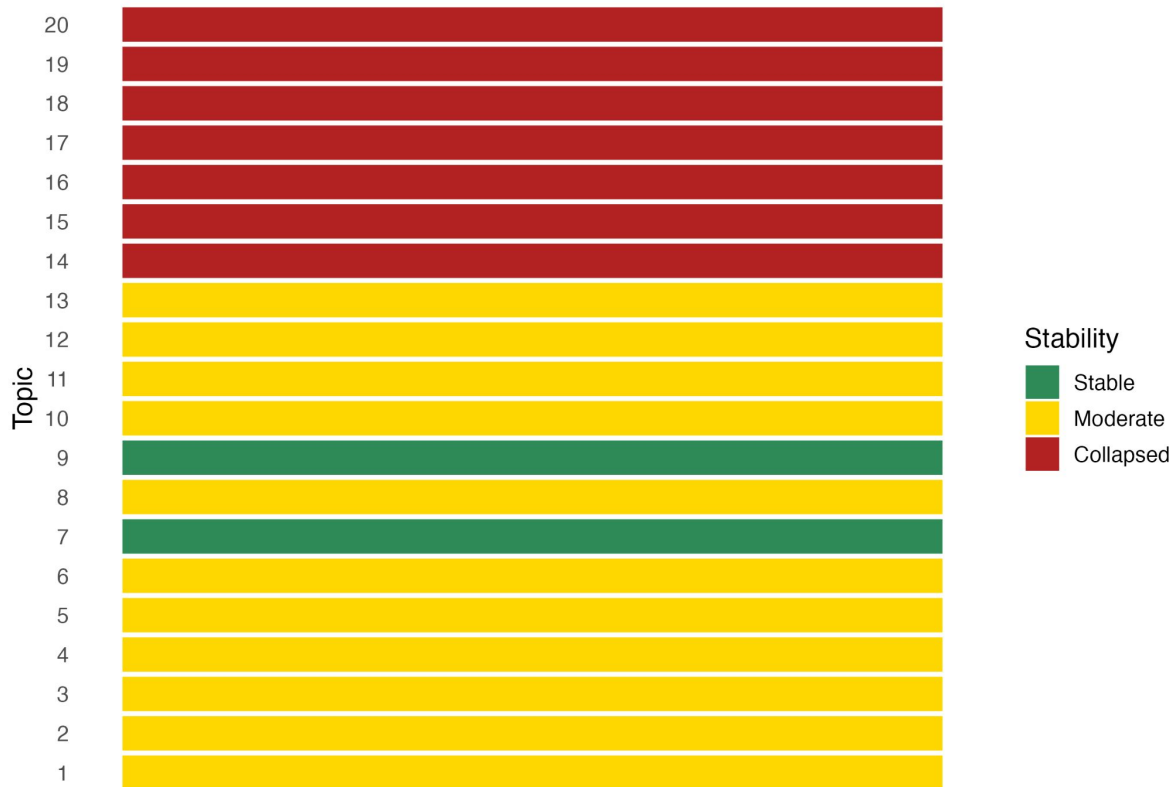
Assume that addition/removal on one document in corpus can only change the topic-word count by one.



??

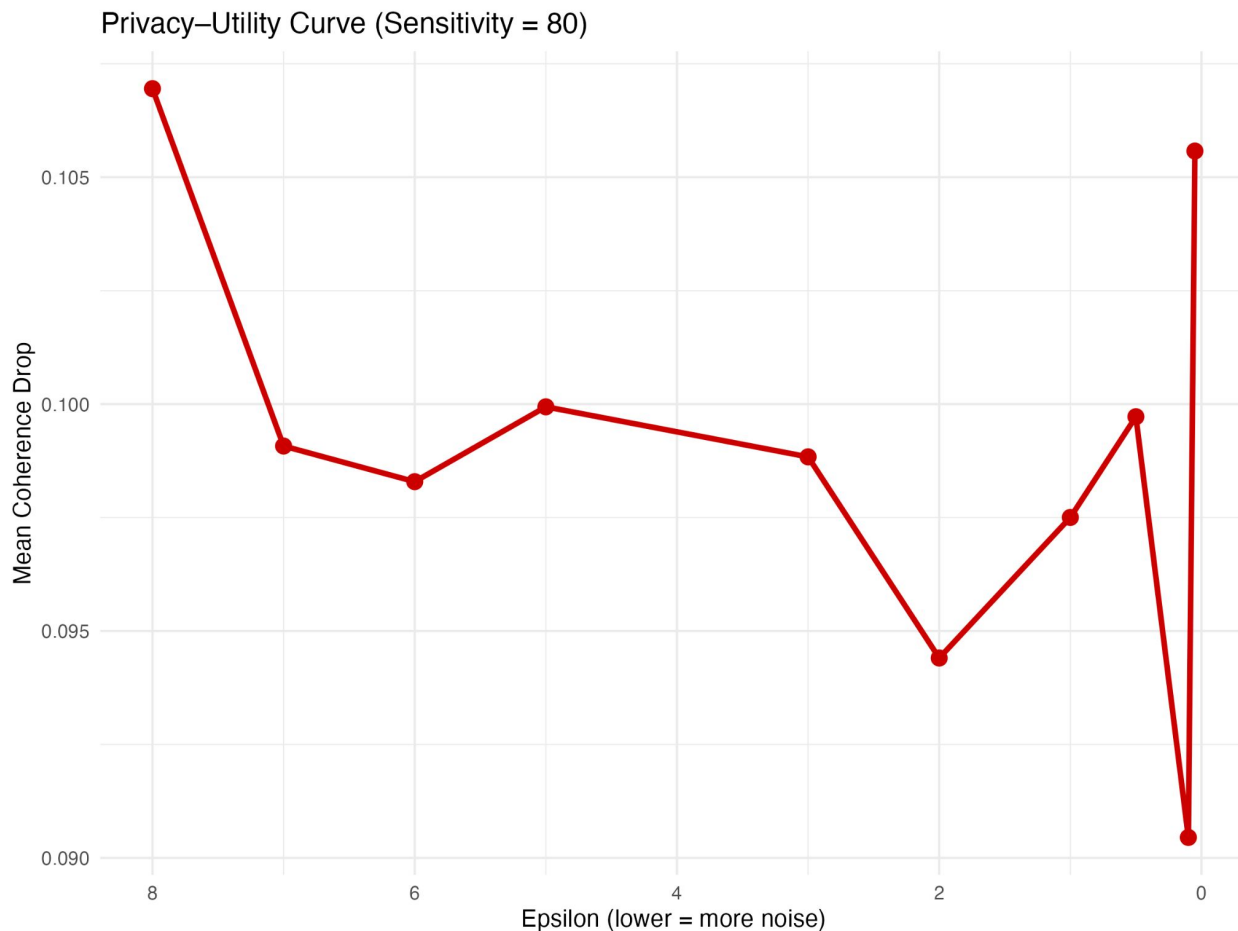
As much as
121 posts!

Topic Stability Heatmap: $\varepsilon = 5$ (less noise) vs $\varepsilon = 0.05$ (high noise)



DP-LDA: Project Approach (User-Level) (III)

Set sensitivity = 80 to mask contribution by outlier user



DP-LDA: Project Approach (User-Level) (IV)

Next Steps: **User Level Privacy**

- Look at the distribution: number of posts by user
- Based on L1 norm histogram, look for a reasonable cut-off (ideally around 10 or as close as to 95th percentile)
- Clip document contributions at max of this cutoff



Questions!

‘Randomness is the protector of secrets!’